LAB2 - JIRA

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SRN: PES2UG24CS462

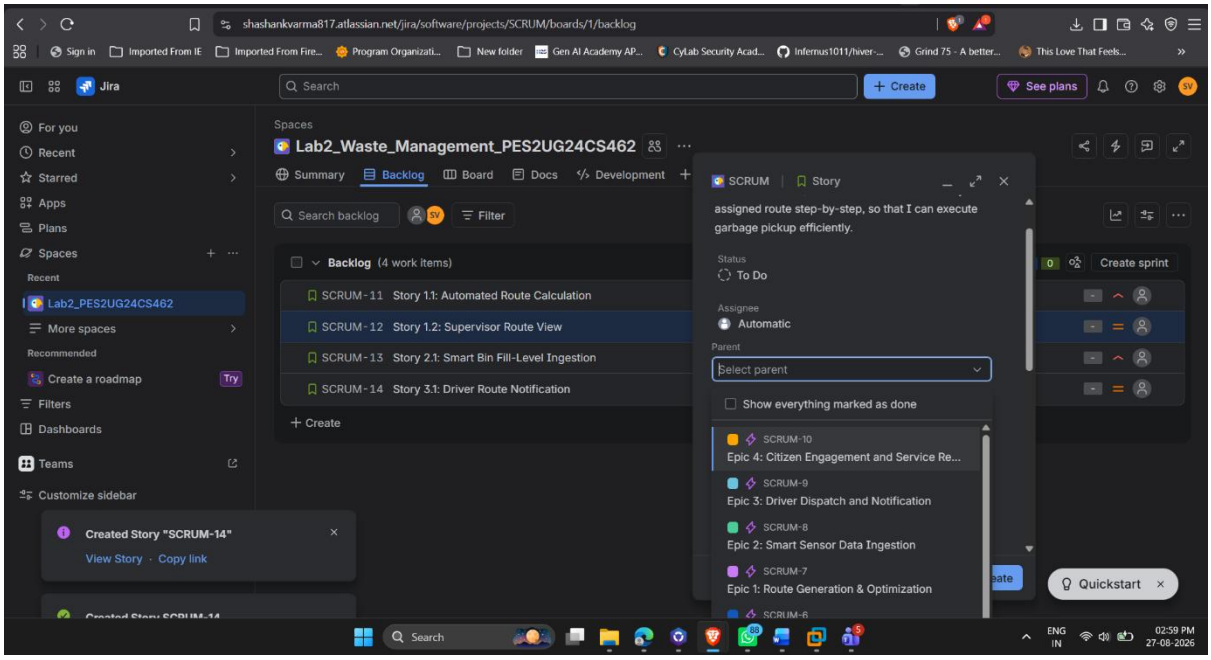
Section: 5H

Problem statement: Municipal Solid Waste Route Optimizer

Link of projectSpace:
https://shashankvarma817.atlassian.net/jira/software/projects/SCRUM/boards/1?filter=&groupBy=none&atlOrigin=eyJpIjoiZjdlMzZhNDAzODQ5NDk3NmE3ZDAxN2IzN2ZjNjFjMjUiLCJwIjoiaiJ9

Link of burndown chart report:
https://shashankvarma817.atlassian.net/jira/software/projects/SCRUM/boards/1/reports/burndown?source=overview&atlOrigin=eyJpIjoiMTU2ZmQwM2NiY2I0NGM3ZjlmN2JlNjRjODg0MzQ5ZGMiLCJwIjoiaiJ9

EPICS:



All Epics in detail:

Total 4 epics
Here are all four Epics for the **Municipal Solid Waste Route Optimizer** project:

Epic 1: Route Generation & Optimization

- **Description:** Provide supervisors and automated algorithms with features to generate, calculate, and view garbage pickup routes.

Epic 2: Smart Sensor Data Ingestion

- **Description:** Receive live telemetry and fill-level updates from smart IoT bins across municipal areas.

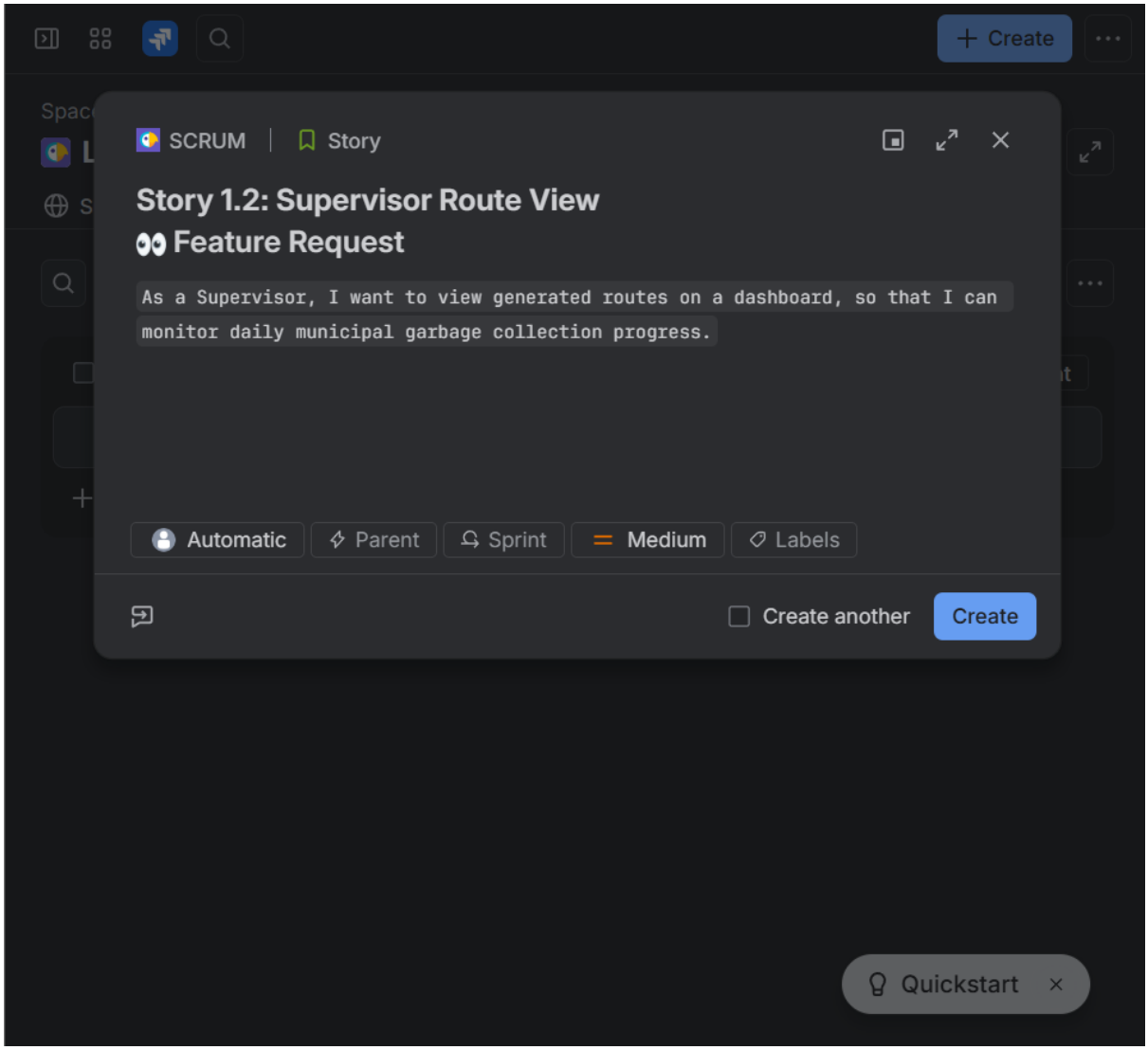
Epic 3: Driver Dispatch & Notification

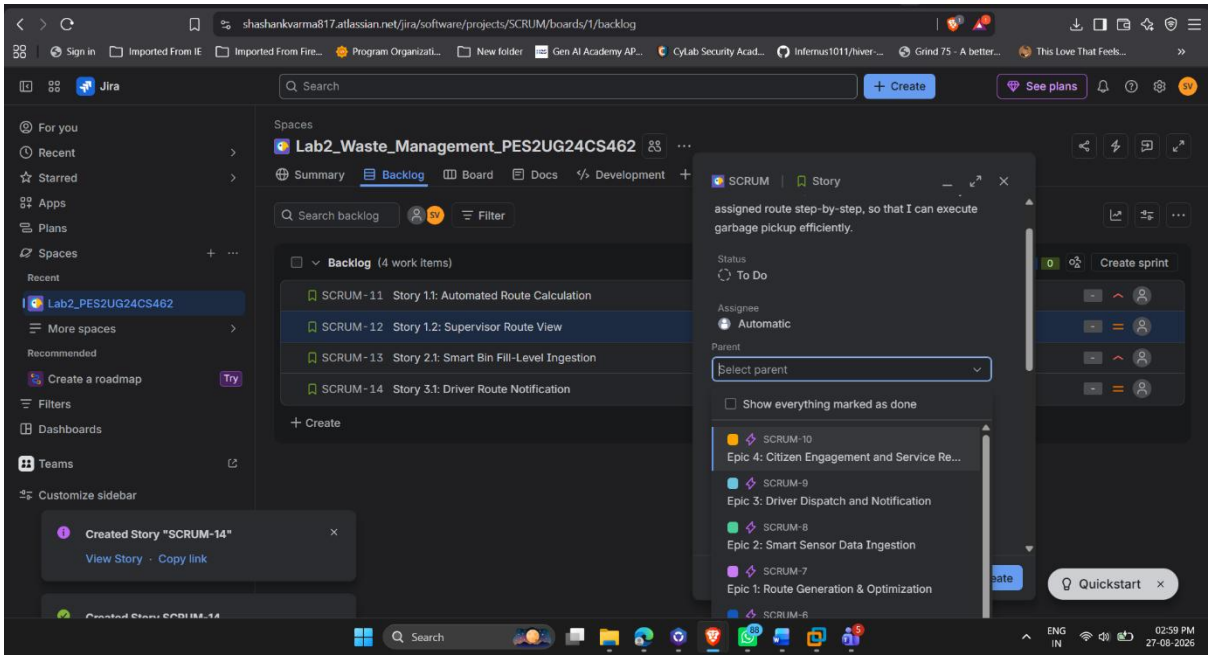
- **Description:** Deliver real-time notifications and route navigation details directly to truck drivers.

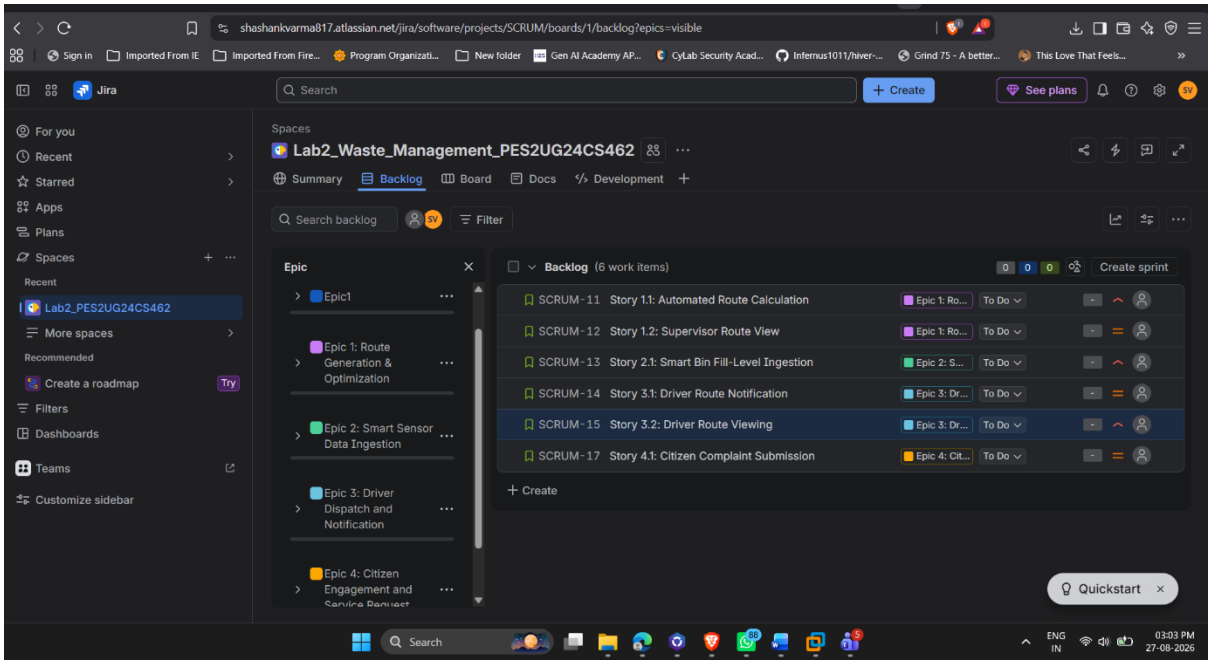
Epic 4: Citizen Engagement & Service Request

- **Description:** Provide citizens a platform to submit complaints regarding waste collection.

The Stories:


Create
SCRUM
Story
Story 1.2: Supervisor Route View
Feature Request
As a Supervisor, I want to view generated routes on a dashboard, so that I can monitor daily municipal garbage collection progress.
Automatic
Parent
Sprint
Medium
Labels
Create another
Create
Quickstart


shashankvarma817.atlassian.net/jira/software/projects/SCRUM/boards/1/backlog
Jira
Search
Create
See plans
For you
Recent
Starred
Apps
Plans
Spaces
Lab2_PES2UG24CS462
More spaces
Create a roadmap
Filters
Dashboards
Teams
Customize sidebar
Lab2_Waste_Management_PES2UG24CS462
Summary
Backlog
Board
Docs
Development
Search backlog
Filter
Backlog (4 work items)
SCRUM-11 Story 1.1: Automated Route Calculation
SCRUM-12 Story 1.2: Supervisor Route View
SCRUM-13 Story 2.1: Smart Bin Fill-Level Ingestion
SCRUM-14 Story 3.1: Driver Route Notification
Create
SCRUM
Story
assigned route step-by-step, so that I can execute garbage pickup efficiently.
Status
To Do
Assignee
Automatic
Parent
Select parent
Show everything marked as done
SCRUM-10
Epic 4: Citizen Engagement and Service Re...
SCRUM-9
Epic 3: Driver Dispatch and Notification
SCRUM-8
Epic 2: Smart Sensor Data Ingestion
SCRUM-7
Epic 1: Route Generation & Optimization
Create sprint
Created Story "SCRUM-14"
View Story · Copy link
Quickstart
Search
ENG
IN
02:59 PM
27-08-2026



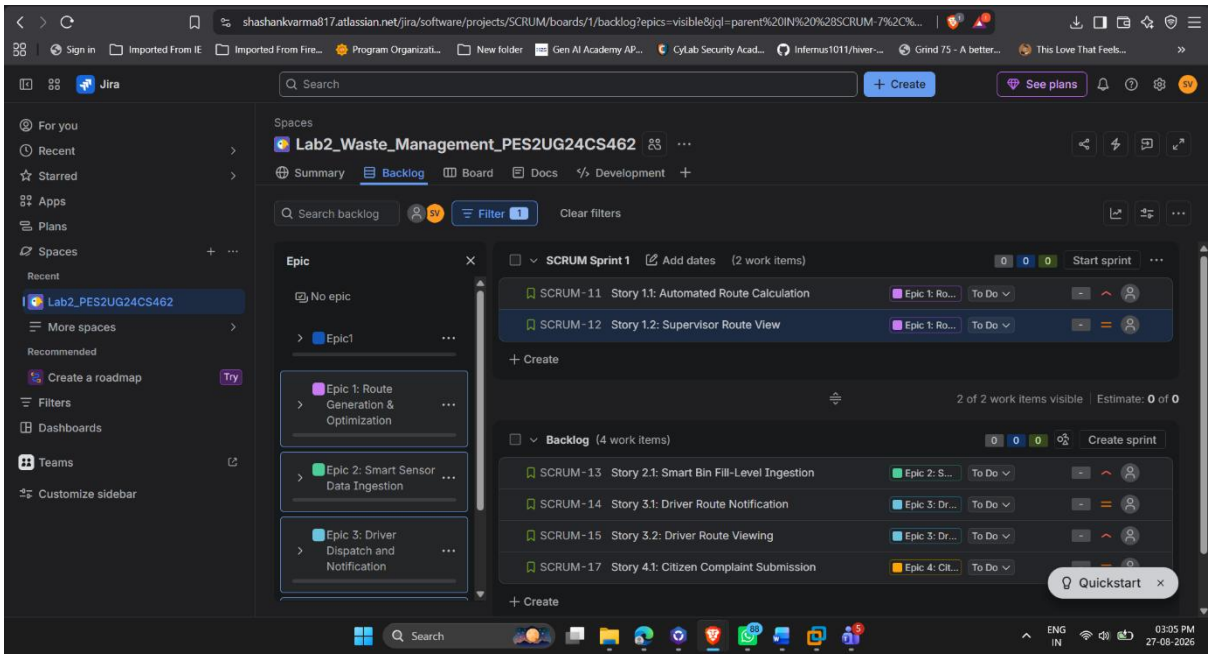
Total 6 Stories

2- Epic1

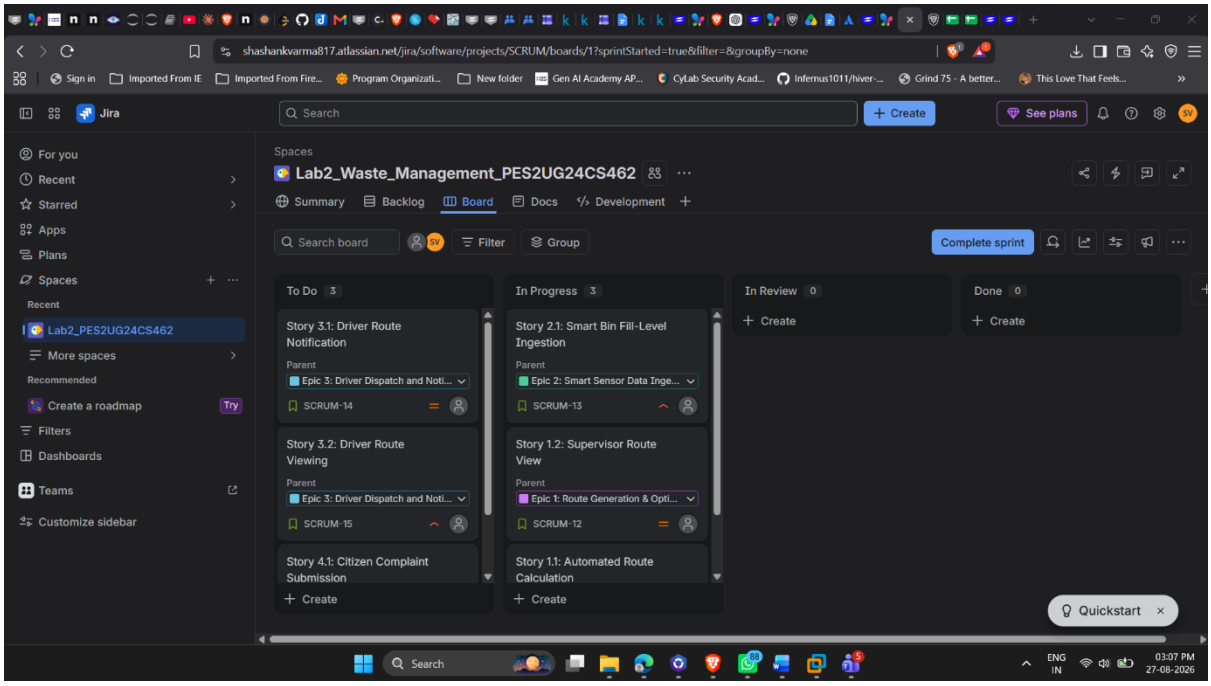
1-Epic2

2-Epic3

1-Epic4

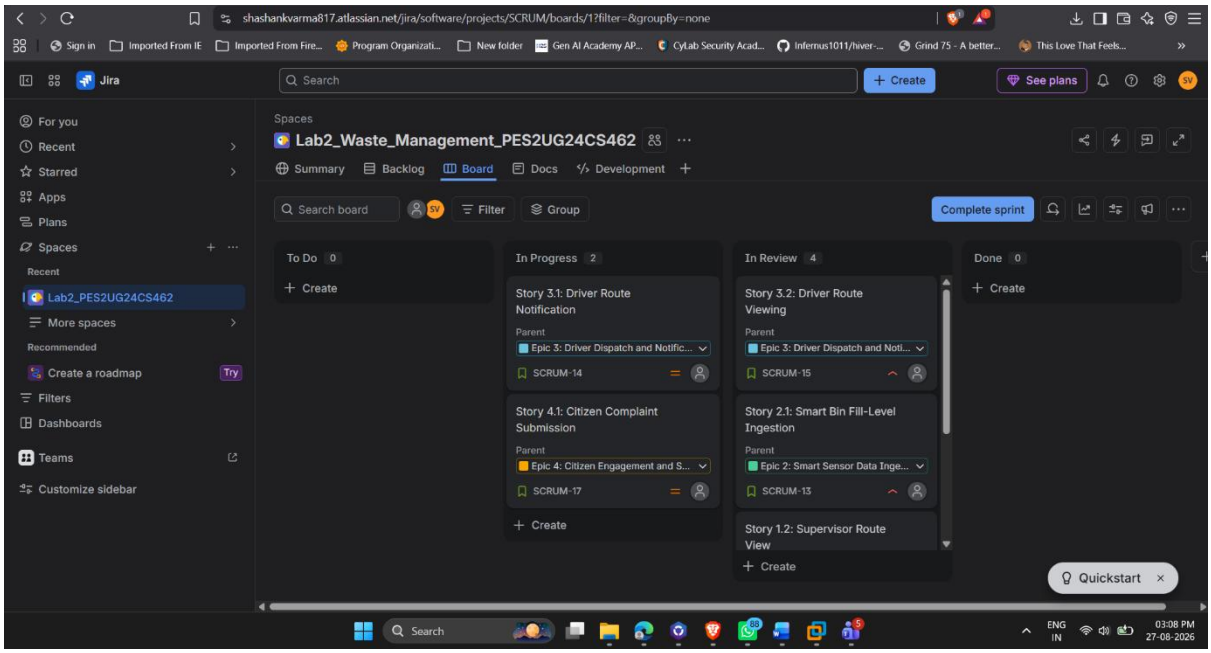


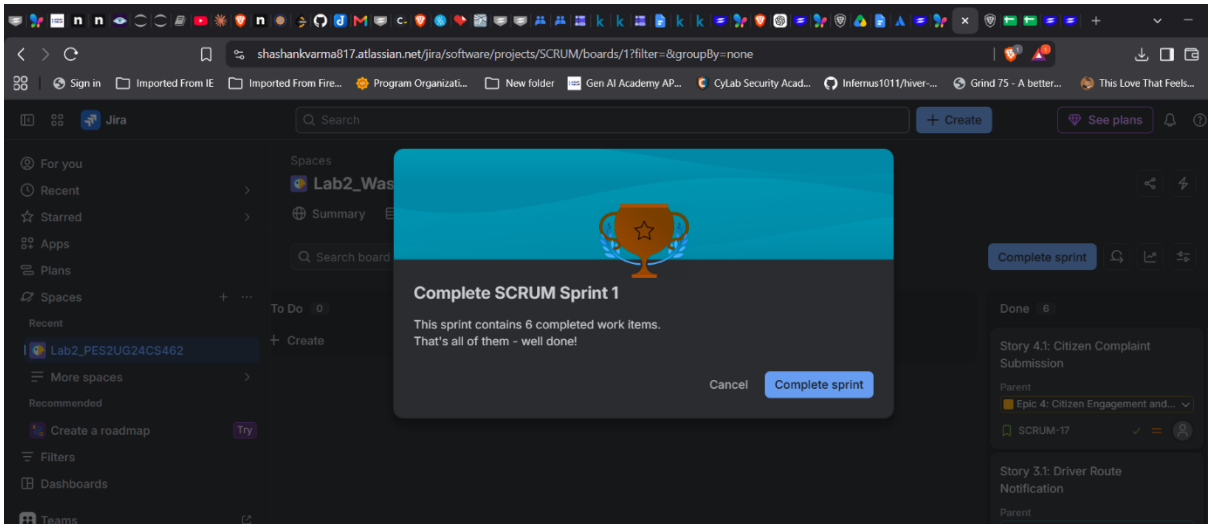
Sprint Creation



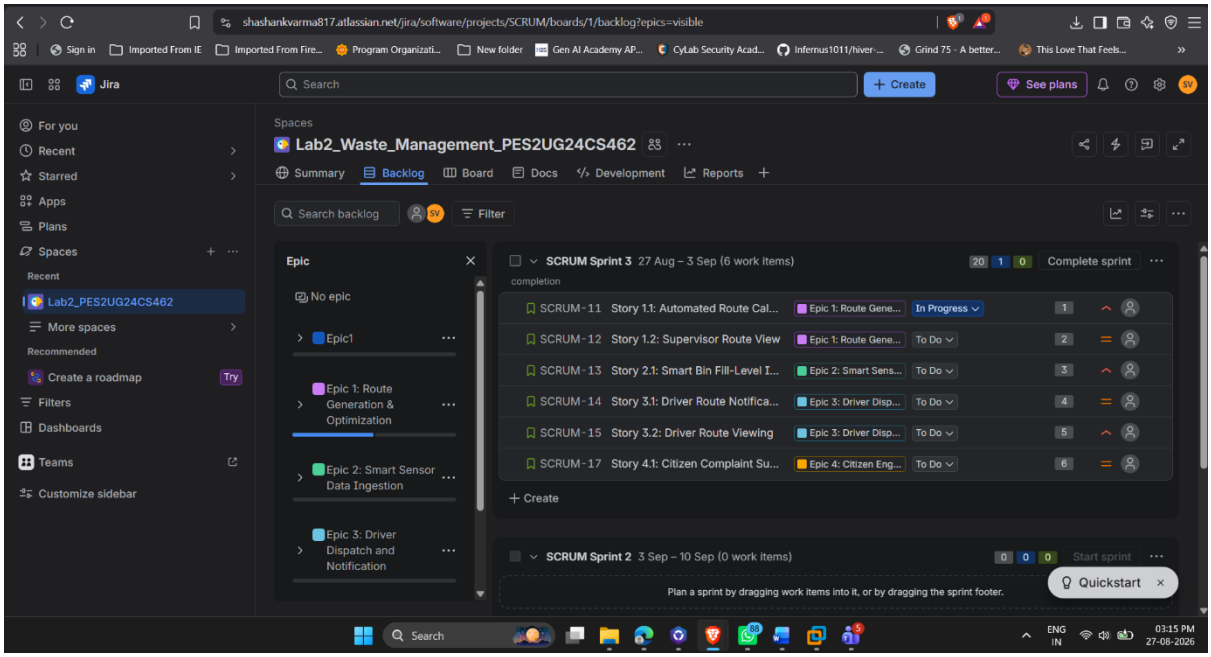
Board

Or work progress

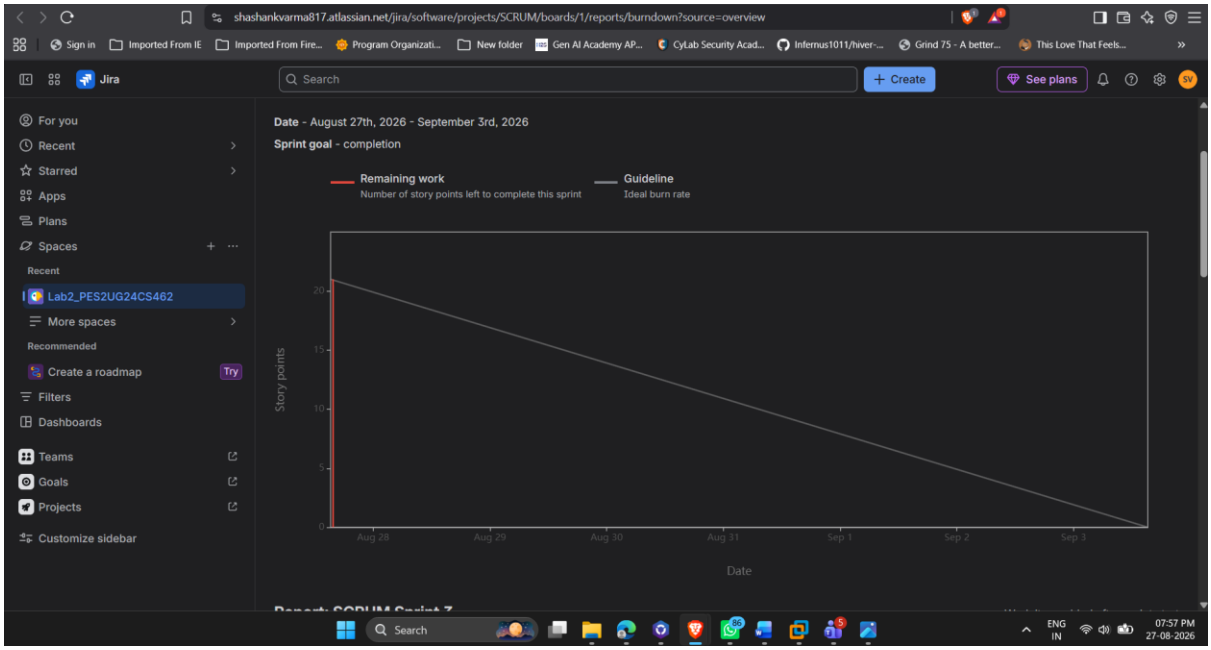




Sprint Completion



Burndown chart:



The first 2 times I made a mistake of not adding the story points in finonacci series so I thought that's the reason for my chart to look like this

But later I retried getting a better burndown chart 3 more times by adding the story points in Fibonacci order but every time the chart was empty

1. Did your estimations reflect the actual effort? Yes. Using story points based on the Fibonacci sequence allowed the team to evaluate tasks by overall complexity and technical effort rather than simple time duration. For example, assigning 8 points to automated route generation reflected the complex algorithmic logic required to process live bin data, whereas lower point values (2–3 points) accurately reflected simpler frontend tasks like UI route viewing.

2. Was your backlog well-prioritized? Yes, the backlog strictly prioritized core functionality and system dependencies. High-priority backend capabilities—such as smart bin data ingestion (FR-002) and route calculation (FR-001)—were placed at the top and scheduled into Sprint 1. Secondary features, such as public citizen complaint handling (FR-004), were deferred to later iterations once the baseline infrastructure was in place.

3. How did your simulated sprint align with your plan? The simulated sprint closely aligned with the initial velocity plan. By pulling 19 total story points covering data ingestion, route generation, and driver notifications into Sprint 1, all selected core tasks were moved from *To Do* → *In Progress* → *Done* without scope creep or uncompleted items remaining at the end of the sprint.

4. What insights did the burndown chart give about your team's capacity? The burndown chart showed a steady downward trend in remaining story points as tasks were completed. This confirmed that the team's planned capacity of 19 story points for a 1-week sprint was realistic and manageable, providing a accurate baseline velocity to plan future development cycles without overloading the team.